Deep Learning Systems Analysis Report

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Report Overview

This project implements a convolutional neural network for handwritten digit image classification. The dataset is the public scikit-learn digits benchmark: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html. I trained a baseline CNN and exactly one experimental variant with dropout to compare a controlled architectural change.

Dataset and Task Description

The dataset contains 1797 grayscale digit images represented by 64 pixel features each. Each sample is an 8 by 8 image labeled from 0 to 9, making it appropriate for a small CNN classification task. The dataset is public, non-synthetic, and not reused from the previous statistical or supervised ML projects in this capstone. Its compact size makes it reproducible on a laptop, but it is simpler than real image systems.

Model Architecture and Design Decisions

The baseline model uses two convolutional layers with ReLU activations, one max-pooling step, flattening, and a final linear classifier. Convolutions are appropriate because nearby pixels carry local shape information, and shared filters reduce the number of learned parameters compared with a fully connected image model. PyTorch was used because it supports imperative, debuggable neural-network development (Paszke et al., 2019).

Experimental Comparison

The single experiment was: Added dropout=0.25 before final linear layer. This comparison is meaningful because dropout is a common regularization technique that can reduce reliance on specific hidden activations, though its value depends on dataset size and overfitting risk. All other major settings were held constant: data split, optimizer, learning rate, batch size, and number of epochs.

Results and Interpretation

After 8 epochs, the baseline CNN reached test accuracy 0.977, while the dropout CNN reached 0.972. The final average training losses were 0.0461 and 0.0850, respectively. Figure 2 compares test accuracy by epoch, and Figure 3 shows the experimental model confusion matrix. The result shows that dropout was competitive but not automatically superior on this small benchmark.

Limitations and Risks

The digits dataset is small, low resolution, and curated, so it does not capture lighting variation, blur, handwriting diversity, or operational image noise. The evaluation uses a single split and a short training schedule. A real deep learning system would need broader validation, calibration, failure-case review, and monitoring for distribution drift.

Ethical and Responsible Use

A digit classifier seems low risk, but the system pattern could be adapted to documents, identity checks, or clinical images where mistakes may affect people. Responsible deployment would require transparency about model limitations, human review for uncertain predictions, and data governance aligned with an AI risk-management framework such as NIST AI RMF 1.0 (NIST, 2023).

Future Improvements

Future work should add cross-validation or repeated splits, evaluate calibration, test robustness against shifted or noisy images, and compare a more realistic validation set. If the system were adapted for healthcare operations, the visual model would need domain-specific images and clinical validation before deployment.

References

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., et al. (2019). PyTorch: An imperative style, high-performance deep learning library. Advances in Neural Information Processing Systems. https://papers.nips.cc/paper/9015-pytorch-an-imperative-style-high-performance-deep-learning-library

NIST. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). https://doi.org/10.6028/NIST.AI.100-1

Scikit-learn developers. Digits dataset documentation. https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html

Selected Figures

Figure 1. Sample digit images.

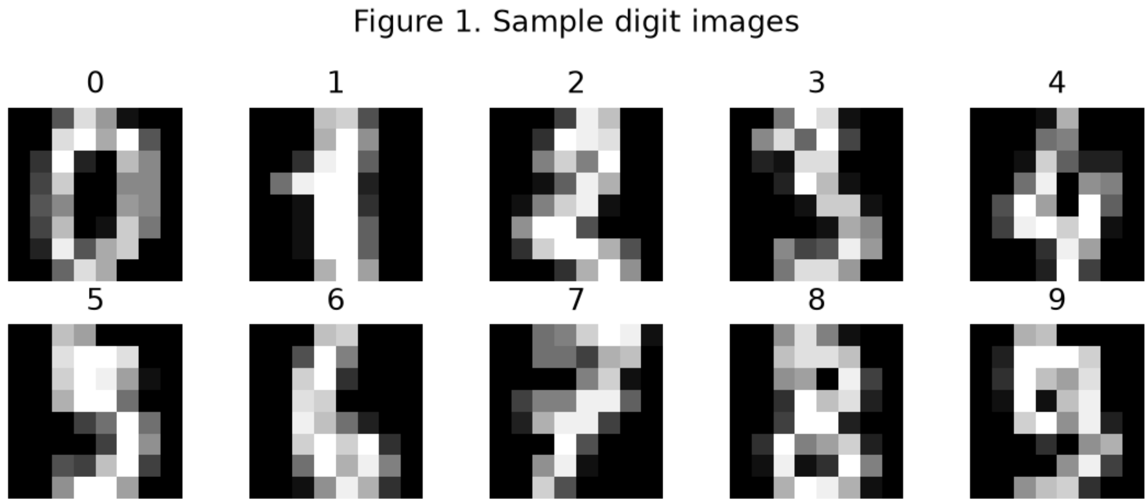

Figure 2. Test accuracy by epoch.

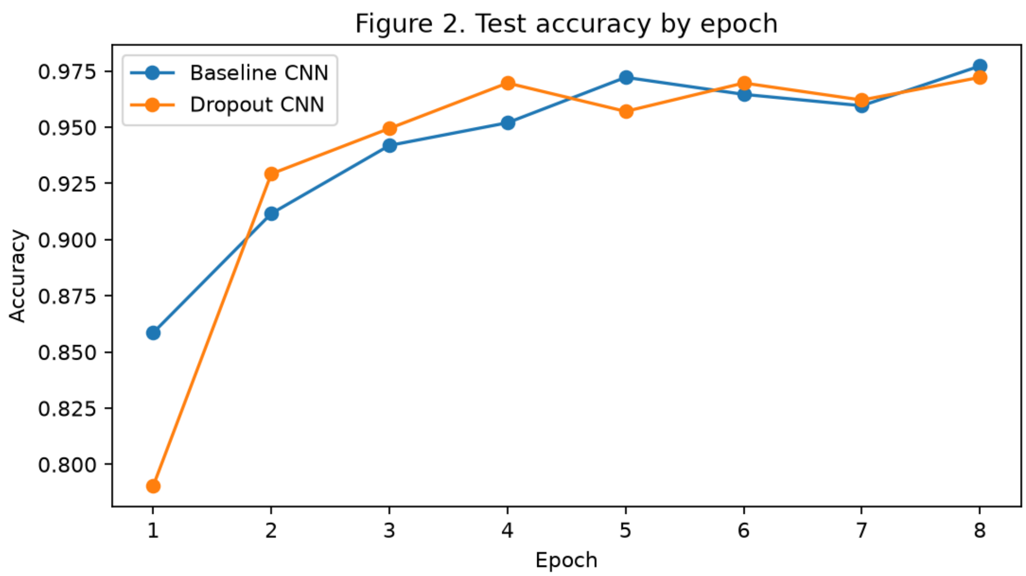

Figure 3. Experimental CNN confusion matrix.

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